

Large Language Models for Generative Recommendation and User Behavior Modeling

Abstract

Large language models have shifted recommendation research from a predominantly discriminative paradigm toward a family of semantic, generative, and increasingly agentic systems. In the classical formulation, recommender systems estimate relevance scores over fixed candidate sets using collaborative filtering, click-through-rate prediction, or sequential user behavior modeling; in the newer formulation, LLMs are used to encode user-item semantics, translate recommendation into language generation, compress heterogeneous behavioral traces into natural-language profiles or semantic tokens, and sometimes directly generate item identifiers, explanations, or interactive responses. This transition matters because recommendation is fundamentally a problem of user behavior modeling under sparsity, heterogeneity, temporal drift, and deployment constraints. LLMs help precisely where traditional pipelines are weakest: leveraging open-world semantics, unifying multiple tasks in a text-to-text interface, and enabling richer interaction with users. At the same time, they introduce new difficulties involving calibration, positional sensitivity, privacy leakage, fairness, efficiency, evaluation, and the mismatch between token prediction and preference learning [1–12]. ¹

This survey synthesizes the literature at the intersection of LLM-based recommendation and user behavior modeling. It first situates the area in the longer history of collaborative filtering, CTR prediction, and sequential recommendation. It then organizes recent work into method families: feature- and representation-augmented recommenders, prompting and frozen-LLM recommenders, fine-tuned and alignment-based LLM recommenders, fully generative recommenders, and LLM-based user behavior modeling systems including simulators and agentic recommenders. The main argument of the survey is that LLMs should not be viewed merely as stronger text encoders. Their deeper significance lies in changing the unit of modeling in recommendation—from latent vectors over user-item matrices to language-grounded, behavior-aware, structurally flexible interfaces that can jointly model preference, content, interaction, and explanation. Yet whether this shift improves real recommendation quality depends critically on how collaborative signals, long-term behavioral structure, and industrial constraints are integrated into the LLM stack [13–130]. ²

Introduction

Recommendation research has historically progressed through several overlapping regimes. Early systems centered on memory-based collaborative filtering and latent factor models, which offered scalable personalization but depended heavily on interaction density and relatively static preference assumptions [1–5]. Deep-learning-era recommenders expanded this foundation with neural matching, feature interaction modeling, and industrial user behavior modeling, especially in click-through-rate prediction and sequential recommendation [6–18]. A third regime then emerged around sequence encoders such as RNNs, CNNs, self-attention, and bidirectional masked models, which treated recommendation as temporal behavior modeling rather than static matrix completion [19–33]. More recent work adds a fourth regime: the language-modeling and foundation-model regime, in which textual side information, prompt-conditioned inference, and generative decoding reshape the recommendation pipeline itself [34–62]. ³

Two concepts in the current literature are often conflated but should be distinguished. The first is **LLM-enhanced recommendation**, where a language model provides semantic features, profile summaries, prompt-based reasoning, or post-hoc reranking while a conventional recommender still performs the core decision-making. The second is **generative recommendation**, where the model directly produces recommendation outputs—natural-language suggestions, semantic item IDs, ranked item sequences, or unified retrieve-and-rank outputs—under an autoregressive or otherwise generative objective [31–33,49–62,93–106]. The distinction is crucial because it determines whether LLMs are acting as auxiliary semantic infrastructure or as the recommender itself. Surveys published from 2023 to 2026 consistently show that feature-based, generative, and agentic paradigms differ not only in architecture but also in problem formulation, evaluation burden, and industrial feasibility. ⁴

User behavior modeling is the natural bridge between classical recommendation and LLM-based recommendation. In industrial systems, behavior logs are long, noisy, heterogeneous, and temporally irregular. Users click, skip, search, purchase, dwell, share, and review; they interact across scenarios such as homepage feeds, ads, search results, and conversational sessions; and their tastes may be multiple, situational, or drifting. Traditional UBM research therefore focused on target-aware attention, recurrent evolution, multi-interest extraction, long-sequence retrieval, and multi-behavior integration [9–18]. LLM-based systems inherit these same requirements, but now must express them through prompts, context windows, semantic tokenizations, adapter layers, and alignment objectives. This creates a central design question for the field: **how can rich collaborative and temporal user behavior signals be made legible to language models without discarding what made recommender systems effective in the first place?** ⁵

The literature suggests that three research pressures drive current progress. First, semantic insufficiency: purely ID-based models often struggle in cold-start or cross-domain settings, where item descriptions and world knowledge become useful. Second, architectural fragmentation: industrial recommenders are usually cascaded into retrieval, pre-ranking, ranking, calibration, and explanation modules, whereas LLMs encourage unification [12,31,49–62,93–106]. Third, interactional expansion: recommendation increasingly involves dialogue, explanation, simulation, human feedback, and open-ended preference communication rather than one-shot top-K prediction [46–48,74–75,89–92,123–130]. The promise of LLMs lies in addressing all three simultaneously, but the evidence also shows meaningful trade-offs in latency, controllability, and reliability. ⁶

Methods

Classical foundations for behavior modeling before LLMs

The pre-LLM foundation remains indispensable. Collaborative filtering and matrix factorization established the core inductive bias that user preference can be learned from relational structure among interactions rather than explicit semantics alone [1–5]. Later neural recommenders extended this idea by replacing hand-designed similarity functions with learned interaction operators, from neural collaborative filtering and Wide & Deep / DeepFM families to specialized industrial CTR architectures [6–10]. These models solved real production problems—feature crosses, sparse IDs, calibration, and high-throughput inference—but generally represented user behavior either as a static embedding or as a shallow aggregation over historical events. They were powerful as ranking engines, less so as general user models.

User behavior modeling research addressed this limitation directly. DIN and DIEN introduced target-aware activation and dynamic interest evolution, making recommendation conditional on how different parts of behavior history relate to the target item [9–10]. SIM, BST, MIND, ComiRec, TWIN, and retrieval-

based UBM models extended the approach to ultra-long or multi-interest histories by combining search, transformer encoding, behavior retrieval, and multi-vector user representations [13–18]. The conceptual insight was that a user is not a single point in latent space but a moving bundle of interests, some persistent and some situational. This insight remains central in LLM-based recommendation, albeit now expressed through prompts, semantic clusters, or persona abstractions rather than only learned embeddings. 7

Sequential recommendation provided a second pillar. GRU4Rec, HRNN, Caser, NARM, STAMP, SASRec, BERT4Rec, TiSASRec, S3-Rec, CL4SRec, DuoRec, and FMLP-Rec progressively sharpened the modeling of order, recency, periodicity, self-supervision, and representation quality in user behavior sequences [19–30]. Much of today’s generative recommendation literature still uses these models as baselines or teachers. This is not an accident: the core tasks of next-item prediction, temporal dependency tracking, and long-range preference abstraction were already well defined before LLMs. What LLMs add is not the existence of sequential modeling, but a new interface for representing what sequential signals mean semantically and how they can transfer across tasks, modalities, or domains. 8

Generative retrieval and diffusion-based recommenders also matter as immediate precursors to LLM-based generative recommendation. Rajput et al. reframed recommendation as generative retrieval over item identifiers [31]. DiffuRec and DreamRec explored non-autoregressive or diffusion-style generation in sequential recommendation, emphasizing distributional item representations and “oracle” preference generation [32–33]. These studies demonstrated that recommendation need not be restricted to scoring fixed candidates. They also exposed a key difficulty that persists in LLM approaches: when items are generated rather than scored, the design of item identities, decoding strategies, and evaluation protocols becomes a first-order problem. 9

Language-model adaptation for recommendation

The first wave of language-model-based recommendation did not yet rely on frontier LLMs. Instead, it adapted pretraining ideas from BERT, RoBERTa, and GPT-style language modeling to recommender tasks [39–45]. P5 was especially influential because it proposed a unified text-to-text perspective in which user-item interactions, metadata, and multiple downstream tasks could all be serialized as language and optimized with a shared generation objective [43]. Zero-shot recommendation as language modeling showed that, under carefully designed prompts, pretrained language models already encode some useful item-level affinity without recommendation-specific training [44]. M6-Rec and related work extended the idea toward multi-modality and unified sequencing [45]. 10

This transitional literature accomplished three things. First, it legitimized the idea that recommendation can be rephrased as language processing rather than only matrix factorization. Second, it exposed the strength of item and user text as a transfer medium across tasks. Third, it clarified the limitations of naïve textification: prompts are brittle, collaborative signals are weakly represented in raw text, and long histories exceed efficient context usage. These issues strongly influenced the next wave of LLM-based work. 11

Conversational and explainable recommendation also formed a natural substrate for LLM adoption. Deep conversational recommendation, neural attentive explanation, and BERT-based probing for conversational recommendation all anticipated the later shift toward language-mediated recommendation [46–48]. Once LLMs became strong enough to sustain multi-turn dialogue and explanation, conversational recommender systems ceased to be a niche subfield and instead became a major use case for LLM-based personalization. The interaction interface moved from selecting among candidate clarifying questions to synthesizing conversations, rationales, and preference state updates in natural language. 12

Feature- and representation-augmented LLM recommendation

A large segment of the literature uses LLMs not as end-to-end recommenders but as high-capacity semantic encoders. The basic pattern is to enrich item or user representations with generated descriptions, inferred attributes, profile summaries, knowledge expansions, or graph-aware textualizations, and then feed those features into otherwise conventional recommenders [49–55,60,67–71,83–89]. In LLM-Rec, prompting is used to enrich textual representations for personalized recommendation [60]. LLMRec incorporates graph augmentation to encode collaborative structure alongside semantics [70]. Representation-learning approaches inject LLM outputs into item or user embeddings, adapters, or two-tower architectures [71,84–89]. The practical appeal of this family is that it preserves the maturity, controllability, and speed of discriminative recommenders while adding open-world knowledge and stronger text understanding. ¹³

Several subfamilies are visible. One focuses on **item-side semantic enrichment**, using LLMs to rewrite sparse titles, infer aspect terms, generate summaries, or add world knowledge. This helps particularly in cold-start and text-rich settings. Another focuses on **user-side profile induction**, converting interaction histories into compact natural-language or pseudo-natural-language summaries that are more interpretable and sometimes more transferable across domains. A third seeks **semantic-collaborative fusion**, where LLM embeddings are aligned with graph or sequence embeddings, often through adapters or contrastive losses [70–71,81,84–89]. The unifying idea is that semantics and collaboration are complementary but not interchangeable. Good systems therefore avoid simply replacing collaborative features with language features; they instead construct interfaces between them. ¹⁴

This design space has produced several consistent findings. Semantics help most in sparse, cold, cross-domain, or text-rich settings. Conversely, in dense warm-start recommendation, pure collaborative baselines often remain very competitive, and sometimes superior, because they are optimized exactly for preference discrimination rather than textual plausibility [31,36,44,61–66]. This helps explain why many strong recent systems are hybrid: collaborative filtering supplies precise preference priors, while LLMs contribute semantics, world knowledge, and language-grounded abstraction. ¹⁵

Prompting, fine-tuning, and alignment-based LLM recommenders

The next major family uses LLMs more directly, either by prompting frozen models or by fine-tuning them for recommendation. Early prompting papers treated recommendation as instruction following, rank completion, or constrained generation over item descriptions or candidate sets [58–68,72–80]. GenRec explicitly fine-tuned an LLM to generate target items rather than score them [58]. PALR combined candidate retrieval with LLM ranking [59]. LLM-Rec showed that prompt-based semantic enrichment could close much of the gap between simple content models and more complex recommenders [60]. Zero-shot ranker results were especially important because they suggested that large pretrained models already encode enough world knowledge and preference-like associations to perform competitive recommendation in low-data settings [61–66]. ¹⁶

Prompting alone, however, is rarely sufficient. The core issue is objective mismatch: recommendation demands calibrated preference discrimination, while decoder-only LLMs are trained for next-token prediction. Fine-tuning methods therefore try to align LLM latent spaces with recommendation signals. TALLRec uses efficient tuning to adapt LLMs to recommendation [72]. Recommendation-as-instruction-following frameworks translate recommendation tasks into instruction-style templates [73]. Collaborative LLMs, latent-space interpretation methods, and collaborative-semantics integration methods seek to bind item-level collaborative structure to token-level language modeling [80–84]. These studies collectively argue that recommendation performance improves when collaborative knowledge is

not merely appended as context but injected into the optimization target or representation geometry.

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Alignment has become a major theme. The field increasingly borrows post-training ideas from LLM alignment—supervised fine-tuning, preference optimization, and reinforcement learning—but adapts them to recommendation-specific supervision. RosePO is a representative example, designing preference optimization for “helpful and harmless” LLM recommenders while addressing semantic hallucination and popularity bias [122]. More recent work on collaborative tokenization and token-level soft-label alignment goes further, attempting to map collaborative filtering logits into the token space so that next-token updates more directly reflect recommender behavior [111]. The central insight is that recommendation alignment is not identical to general LLM alignment: it must reflect comparative preference structure, catalog-specific constraints, and behavioral uncertainty, not just generic helpfulness. 18

Fully generative recommendation and large recommendation models

Fully generative recommendation reframes the task most aggressively. Instead of predicting scores for candidate items, the model emits item identifiers, semantic ID sequences, or recommendation lists through generation [31,58,77,81,87,93–106]. The appeal is architectural unification: retrieval, ranking, explanation, and even search can in principle be expressed within one generative model. But the approach only became a serious industrial contender after the emergence of large recommendation models such as HSTU. Zhai et al. showed that recommendation can be cast as sequential transduction at very large scale, with strong empirical performance and evidence of scaling behavior analogous to language models [93]. Survey work from 2025–2026 treats this as a watershed moment because it turned generative recommendation from an academic curiosity into a plausible replacement for parts of the traditional cascade. 19

Recent work expands this paradigm in several directions. Multi-Behavior Generative Recommendation extends generation from single-type interaction histories to diverse behavior signals [94]. AtSpeed addresses inference cost through speculative decoding for top-K generative recommendation [95]. Universal recommendation learners aim to unify scenarios, objectives, and even search–recommendation tasks within a single framework [96]. OneRec, large-scale generative ranking frameworks, industrial end-to-end advertising models, and MTGR push unification further by collapsing retrieve-and-rank boundaries or replacing cascades with one generative stack [97–106]. Parallel semantic ID generation, sparse–dense hybrid generation, and generative pretraining for discriminative recommenders all suggest that the field is converging on a broader idea: recommendation is increasingly treated as **tokenized behavior transduction**, with tokens standing for items, aspects, or semantic codes rather than ordinary words. 20

Still, three unresolved technical questions remain. First, what should the model generate—item text, original item IDs, vector-quantized semantic IDs, or multi-token semantic codes? Second, how should generation be constrained so that outputs are catalog-valid, diverse, and efficient? Third, what are the right training targets when user preference is historically underdetermined and context dependent? The literature has no settled answer. Some papers favor semantic IDs because they expose structure and reduce vocabulary mismatch; others preserve item IDs to maintain precise catalog grounding; still others hybridize dense retrieval and generative decoding [31,93,95,102–105]. The broader significance is that item identity itself has become a modeling choice rather than a fixed primitive. 21

LLMs for user behavior modeling, simulation, and agentic recommendation

The user-modeling dimension of the literature is now as important as the recommender-model dimension. One branch uses LLMs to summarize or structure long histories into compact user profiles, personas, or semantic prompts [71,83–89,108–109]. Another uses LLM agents to simulate behavior trajectories, either for conversational recommendation, cold-start augmentation, offline evaluation, or multi-agent coordination [74–75,89–92,123–130]. These two branches are closely related: once the system can write a user profile in language, it can also role-play or simulate that user in downstream interactions.

User simulation papers are especially revealing about what “user behavior modeling” means in the LLM era. Early agent work showed that LLMs could generate plausibly human interaction sequences in sandbox environments [75]. Recommendation-specific systems such as RecAgent, On Generative Agents in Recommendation, and cold-start simulators adapted this capability to recommender settings [74–75,91–92]. Later work raised the bar by grounding simulators in explicit rating mechanisms, temporally aware profiles, or real human–LLM conversation logs, as in RecUserSim and RealUserSim [123,127]. The field is thus moving from surface plausibility toward behavioral fidelity. This is an important distinction: a simulator that sounds human is not necessarily one that yields faithful preference trajectories or reliable offline evaluation. ²²

Agentic recommendation generalizes this further. In agentic systems, LLMs do not merely encode users or rank items; they plan, request tools, clarify constraints, remember interaction state, and coordinate multiple modules or agents [74,90–91,124–130]. This is promising because many real recommendation tasks are not simple next-item prediction problems. They involve retrieval, explanation, follow-up questioning, conflict resolution among multiple objectives, and adaptation to changing intent. Yet agentic systems also intensify evaluation difficulty: success depends not only on ranking metrics but on dialogue efficiency, planning reliability, tool-use correctness, and robustness to ambiguous or shifting goals. For this reason, the agentic literature should be read not as a replacement for recommender systems, but as an expansion of the recommendation problem into interactive decision support. ²³

Challenges and Prospects

Modeling mismatch, evaluation, and benchmark design

The most basic challenge is a mismatch between LLM training objectives and recommender objectives. Next-token prediction rewards fluent continuation, whereas recommendation requires calibrated preference comparisons under a constrained catalog and uncertain observational data. This mismatch shows up as semantic hallucination, weak collaborative fidelity, and unstable ranking behavior. It also complicates evaluation. Classical top-K metrics such as HR, Recall, NDCG, and MRR remain useful for fixed-catalog recommendation, but become incomplete once systems generate explanations, profiles, semantic IDs, or multi-turn dialogues [11–12,31–33,49–62]. Newer surveys therefore recommend multi-dimensional evaluation spanning ranking quality, generation quality, diversity, fairness, safety, and interaction success. Yet few benchmarks simultaneously cover all of these. ²⁴

A second issue is the inadequacy of many legacy datasets. MovieLens, Amazon, and Yelp were instrumental for earlier recommender research, but they are static, offline, and weakly interactive. They do not natively evaluate multi-turn preference elicitation, proactive assistant behavior, simulation fidelity, or unified search-recommendation generation. Recent surveys therefore argue that generative recommendation needs benchmarks reflecting dynamic user feedback loops, heterogeneous action types, and multi-task settings rather than only one-shot next-item prediction [11–12,34,53–62,123–130].

This is one of the strongest prospects for the field: LLM-based simulation, if carefully validated, may enable richer and cheaper evaluation than static logs alone. But whether simulated users can stand in for real users remains unsettled. ²⁵

Efficiency, controllability, and industrial deployment

The strongest barrier to deployment is efficiency. LLM-based reranking and generation are expensive compared with classical recommenders, especially when long histories, large candidate sets, or beam-search decoding are involved. This has led to several strategies: offloading LLMs to representation generation rather than online inference; using adapters, distillation, or lightweight alignment; semantic-ID generation to shorten outputs; on-device compression; and decoding acceleration through speculative or relaxed verification methods [84–89,95,102,110–114]. Industrial papers from 2024–2026 show that the community is not merely scaling models upward; it is actively redesigning item representations, decoding procedures, and serve-time architectures to make generative recommendation viable under strict latency budgets. ²⁶

Controllability is equally important. In a classical recommender, developers explicitly manage the candidate pool, ranking objective, and business constraints. In an LLM recommender, prompt wording, candidate order, decoding temperature, and hidden semantic priors can all change the outcome. Position bias is now well documented in listwise LLM reranking, and architectural remedies such as permutation-invariant rerankers have only recently appeared [110]. More generally, the field still lacks a mature theory of controllable generative recommendation: one that would let practitioners specify business rules, diversity quotas, fairness constraints, exposure controls, and catalog validity while preserving the model’s semantic flexibility. This is a major open research direction. ²⁷

Fairness, privacy, robustness, and human values

LLM-based recommendation inherits the classic harms of recommender systems—popularity bias, exposure inequality, and representation bias—but adds new ones from foundation models, including prompt sensitivity, pretrained social stereotypes, long-context leakage, and explanation-level bias. Empirical work finds that LLM recommenders can exhibit or amplify popularity and fairness problems, though in some settings prompting and alignment can mitigate them [116–120,122]. What makes the problem harder than in classical recommender systems is that unfairness may arise from multiple layers simultaneously: pretraining corpora, prompt serialization, generated rationales, preference-pair construction, or feedback loops. This has motivated dedicated fairness evaluation frameworks and post-training methods such as RosePO and BiFair. ²⁸

Privacy is also sharper in LLM-empowered recommendation because user histories and profiles are often literally serialized into prompts. Recent work demonstrates reconstruction and inference risks against LLM-based recommenders, showing that linguistic reparameterization does not remove privacy risk and can in some cases make leakage easier to study or exploit [121]. Robustness concerns are similarly broader than classical adversarial recommendation. They now include jailbreak-like prompt manipulation, spurious sensitivity to candidate order, hallucinated explanations, distribution drift across domains, and unreliable user simulation. For recommendation researchers, the main lesson is that trustworthy LLM-based recommendation cannot be achieved by simply importing general LLM safety methods; it requires domain-specific work on exposure, behavior, and personalization. ²⁹

Conclusion

The development of LLM-based recommendation should be understood not as a sudden break with recommender-systems history, but as the latest stage in a much longer attempt to model users from incomplete, heterogeneous, and temporally evolving behavioral evidence. Classical collaborative filtering taught the field how to exploit relational structure. Deep user behavior modeling and sequential recommendation taught it how to encode interest evolution, target awareness, and long histories. The LLM era adds a new capability: the ability to language-ground users, items, contexts, goals, and explanations within one flexible interface [1–33,39–62]. This is why LLMs matter for recommendation even when they do not dominate all leaderboards. They change the representational language of the problem.

At the same time, the evidence does not support a simplistic claim that “LLMs replace recommender systems.” The highest-confidence reading of the literature is more nuanced. In warm-start, high-density settings, classical collaborative and sequential models remain extremely strong. In sparse, cold-start, multi-domain, interactive, or semantics-heavy settings, LLMs provide clear advantages. The most promising systems are therefore hybrid: they combine collaborative preference learning with semantic enrichment, behavior abstraction, alignment, and, in some cases, generative decoding or agentic interaction [55–62,70–71,84–95,108–130]. ³⁰

Three prospects appear especially consequential. First, **unified recommendation engines** may reduce fragmentation across retrieval, ranking, search, explanation, and conversation. Second, **language-grounded user behavior modeling** may yield more expressive and transferable personalization than traditional fixed-vector user embeddings. Third, **simulation and agentic interfaces** may expand recommendation from passive ranking to active decision support. But all three prospects depend on solving foundational issues in evaluation, efficiency, fairness, privacy, and controllability. For that reason, the future of the field is unlikely to be defined by ever-larger models alone. It will be defined by better ways of connecting language modeling with the actual structure of human preference and behavior. The bibliography below is intentionally selective rather than exhaustive, emphasizing directly relevant and high-confidence works across both recommendation and user behavior modeling. ³¹

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